

# Lecture 2: Supply and Demand (Chapter 2)

*Applied Microeconomics — Master Notes · by Jakob V. Stangel*

**Required reading:** Friberg, ch. 2 (*Supply and Demand*). • Lecturer: **Luigi Butera**.

**Theme of the lecture (Butera):** build tractable models of demand and supply, then use them to find the **market equilibrium** — the price at which "no one would benefit from changing their own behaviour given the choices of others." The whole of micro is understanding that equilibrium: what sets prices and quantities, and how they move.

★ **The heart of the chapter:** a market is one picture — a **downward-sloping demand curve** and an **upward-sloping supply curve** that cross at **exactly one point, the equilibrium** ( $p^*, q^*$ ), where **quantity demanded = quantity supplied**. Three skills: ① write demand & supply as **functions**; ② set them equal to **solve for  $p^*$  and  $q^*$** ; ③ **shift** a curve and read off the new equilibrium. *Price is set by BOTH blades of the scissors (Marshall) — never demand or supply alone.*

**The red thread.** For every question ask the one diagnostic that trips people up: **is this a movement ALONG a curve (only the good's own price changed) or a SHIFT of the whole curve (something else changed — income, other prices, costs, tastes, tech)?** Own price → slide along. Anything else → the curve moves. Get that right and the rest is algebra.

## Key concepts / "modes" to use

- **Demand function**  $Q_D = D(p, p_s, p_c, Y, \tau)$  → its 2-D demand curve; **inverse demand** ( $p$  as a function of  $q$ ).
- **Supply function**  $Q_S = S(p, p_o)$  → **supply curve**; **inverse supply**.
- **Movement along vs shift** of a curve (the core distinction); **shifters** of each curve.
- **Substitutes / complements**; **normal / inferior** goods; **Giffen** good (the rare exception).
- **Market equilibrium**  $Q_D = Q_S$  → solve for  $p^*, q^*$ ; **excess demand (shortage)** / **excess supply (surplus)**.
- **Comparative statics** (shifts) & how the **slope** of the other curve sizes the effect.
- **Price ceiling / floor**; **perfect competition** assumptions; **Marshall's scissors**.

## Section-by-section main points

### 2.1 Demand — the demand function

**Core:** demand answers "how much do buyers want at each price?" The **law of demand**: all else equal, **lower price → higher quantity demanded** (the curve slopes down).

- **In plain terms — why it slopes down:** two reasons. ① **Scarcity** — you don't have infinite money, so a lower price lets you afford more. ② **Diminishing marginal utility** — the *first* slice of pizza is worth a lot to you; the 30th almost nothing, so you'll only buy more if it's cheaper.
- **The demand function** describes quantity demanded  $Q_D$  as a function of price **and everything else**:

### DEMAND FUNCTION

$$Q_D = D(p, p_s, p_c, Y, \tau)$$

*Reads:* how much buyers want depends on the good's own price **and** a handful of other factors. —  $Q_D$  = quantity demanded (what we're solving for) •  $D(\cdot)$  = "is a function of" •  $p$  = the good's **own price** •  $p_s$  = price of a **substitute** •  $p_c$  = price of a **complement** •  $Y$  = consumers' **income** •  $\tau$  = **taxes**.

- **How the 2-D curve appears** — "flush" the other variables into a constant. A demand *curve* plots  $Q_D$  against **own price only**, so you **fix** everything else at set numbers; they collapse into the intercept. The book's worked reduction — start from a specific form and plug in temperature  $T=20^\circ$ , substitute price  $p_r=2$ , income  $I=5$ :

### COLLAPSING TO A LINEAR DEMAND CURVE

$$Q_D = 1 - p + 0.25 \cdot T + 0.75 \cdot p_r + 0.5 \cdot I = 1 - p + 5 + 1.5 + 2.5 = 10 - p$$

(intercept  $a = 10$  bundles all the non-price factors)

*Reads:* plug fixed numbers into every non-price factor and they merge into one constant. —  $T$  = temperature ( $=20$ ) •  $p_r$  = price of raspberries ( $=2$ ) •  $I$  = income ( $=5$ ); the **coefficients** (0.25, 0.75, 0.5) say how strongly each shifts demand • adding the constant pieces ( $1 + 5 + 1.5 + 2.5$ ) gives the **intercept  $a = 10$** , so only  $p$  is left as a variable  $\rightarrow Q_D = 10 - p$ .

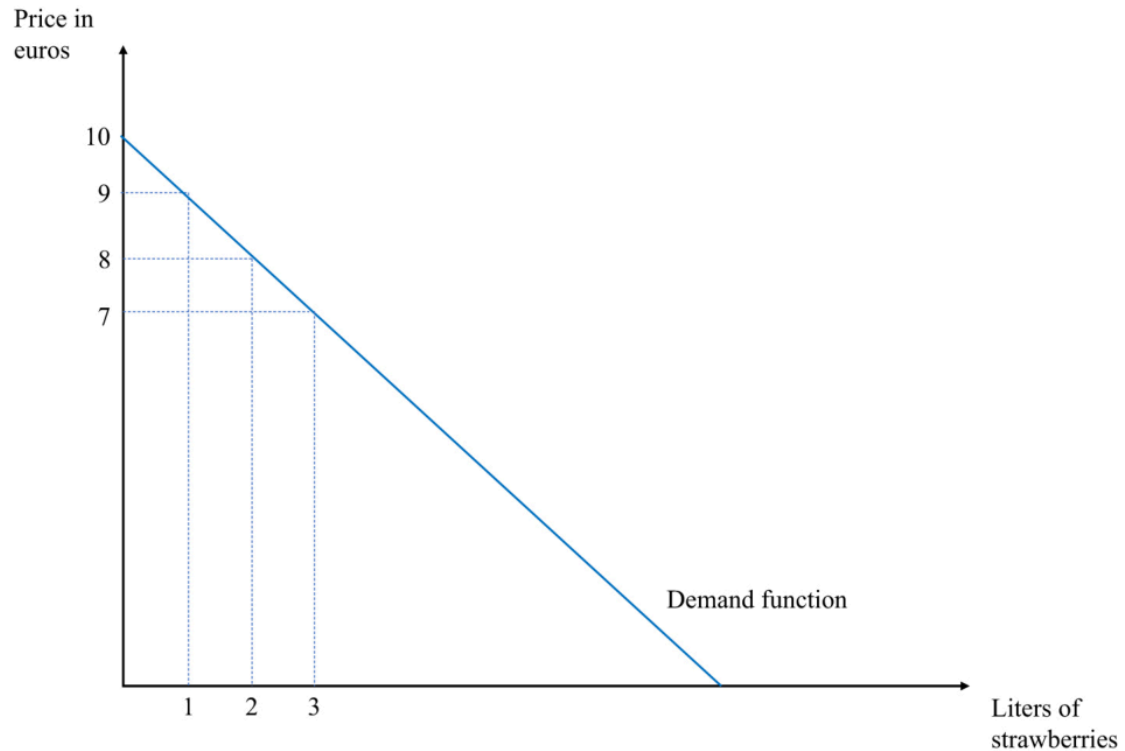
### LINEAR DEMAND & ITS INVERSE (FOR GRAPHING)

$$Q_D = a - b \cdot p \rightarrow \text{e.g. } Q_D = 10 - p \text{ (} a = 10, b = 1 \text{). Solve for } p \Rightarrow \text{inverse demand: } p = 10 - Q_D \cdot (\text{deck's version: } Q_D = 20 - 2p \Rightarrow p = 10 - \frac{1}{2} \cdot Q_D)$$

*Reads:* a straight-line demand — quantity falls as price rises. —  $a$  = the **intercept** (quantity demanded if price were 0; it bundles income, other prices, tastes — *shift it and the whole curve moves*) •  $b$  = the **slope** (how many units demand drops per +1 in price) •  $p$  = price. The **inverse** just rearranges it to put  $p$  alone on the left, because graphs put price on the vertical axis.

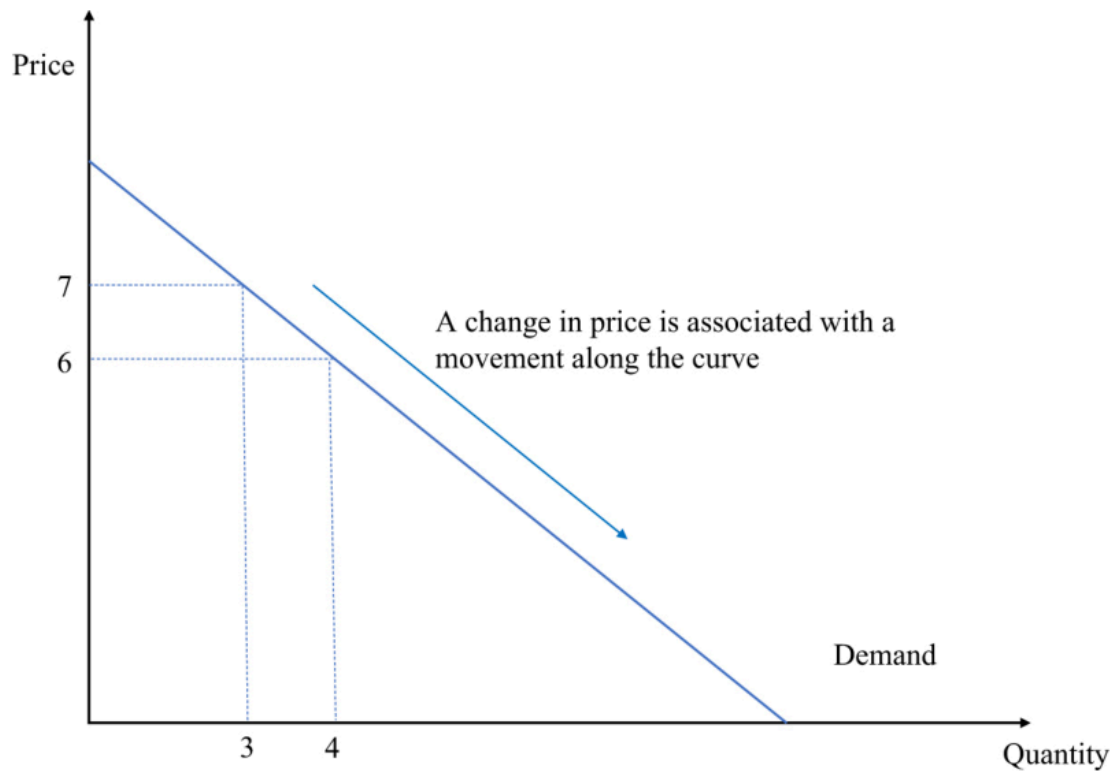
*Check:*  $p = 7 \rightarrow Q_D = 3$ ;  $p = 6 \rightarrow Q_D = 4$ .  $a$  = the intercept (shift it and the whole curve moves);  $b$  = how many units demand falls per +1 in price.

## Supply and demand



**How to read it (the demand curve).** By convention **price is on the vertical axis**, quantity on the **horizontal** (blame Alfred Marshall). Each point answers *both* questions: "at price  $p$ , how much is bought?" *and* "for quantity  $q$ , what's the most a buyer will pay?" Here the line hits  **$p = 10$  when  $q = 0$**  (nobody buys above €10) and slides down to more litres as price falls.

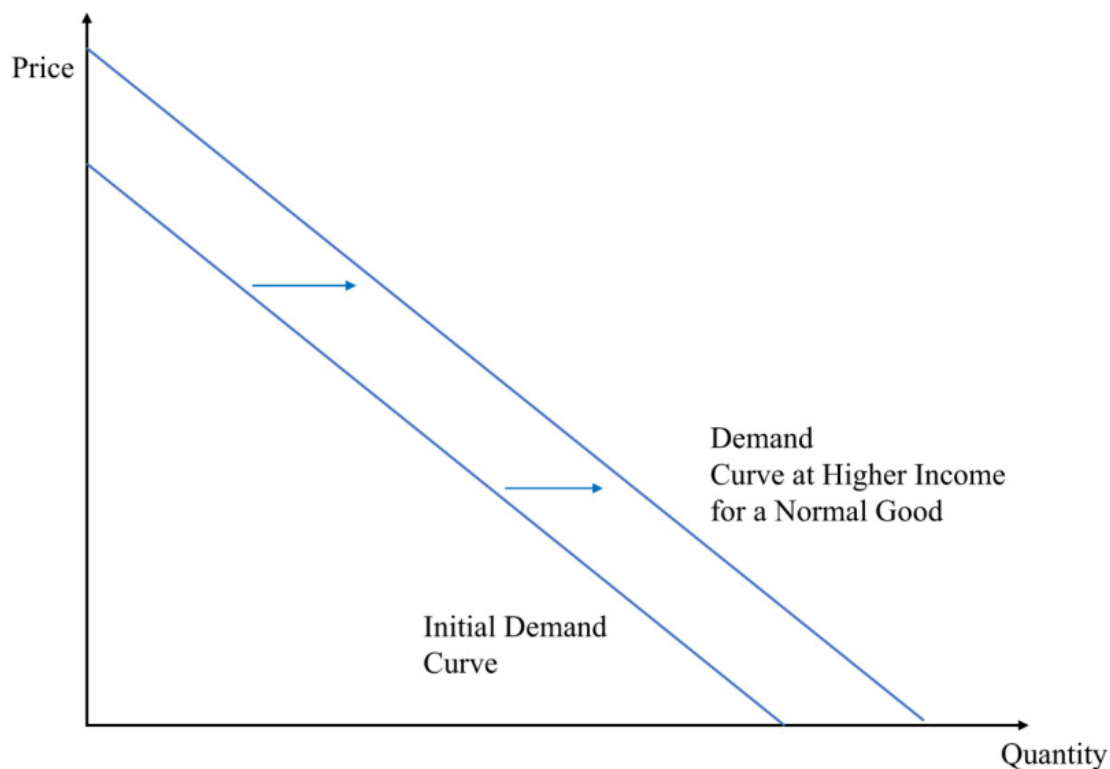
- ★ **Movement ALONG vs SHIFT — the make-or-break distinction:**
  - Own price changes → **MOVE ALONG** the existing curve (you slide to a new point).
  - Anything else changes → the whole curve **SHIFTS** (a *new* curve).



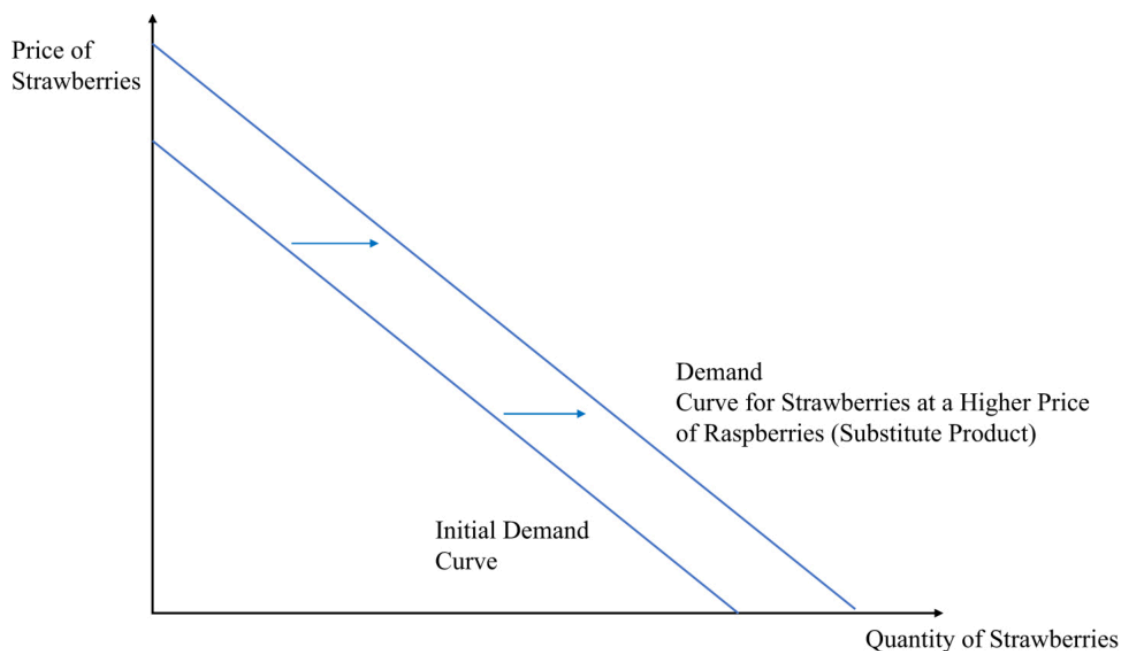
**How to read it (movement along).** Only the **own price** moved (7→6), so you **stay on the same curve** and just slide to a new point (q 3→4). No new curve is drawn. Contrast the shifts below, where the curve itself jumps.

- **Demand shifters** (each moves the *whole* curve = changes the intercept a):

| Shifter                                                                   | Direction                | Why                                           |
|---------------------------------------------------------------------------|--------------------------|-----------------------------------------------|
| <b>Income</b> ↑ (normal good)                                             | shift <b>out</b> (right) | richer → buy more at every price              |
| <b>Income</b> ↑ (inferior good, e.g. instant noodles)                     | shift <b>in</b> (left)   | richer → buy <i>less</i> of it                |
| <b>Price of a substitute</b> ↑ (raspberries ↑ → strawberry demand)        | shift <b>out</b>         | switch toward the now-relatively-cheaper good |
| <b>Price of a complement</b> ↑ (cereal ↑ → milk demand)                   | shift <b>in</b>          | the pair is consumed together                 |
| <b>Tastes / expectations</b> (heatwave → ice-cream; COVID → toilet paper) | out or in                | common-sense direction                        |



Microeconomics



**How to read them (shifts).** The whole line moves **right (out)** = more demanded at *every* price. Fig 2.4: higher **income** (normal good). Fig 2.6: a **substitute** (raspberries) got dearer, so buyers swing to strawberries. Note you are *not* sliding along the old curve — a brand-new curve sits to the right.

- **Substitutes vs complements (definitions):** **substitutes** — demand for good 1 **rises** when good 2's price rises (raspberries ↔ strawberries). **Complements** — demand for good 1 **falls** when good 2's price rises (strawberries ↔ raspberries).

price rises (milk  $\leftrightarrow$  cereal).

- **From individual to aggregate demand (deck).** Market demand = **horizontal sum** of every buyer's demand — *add the quantities at each price*:

#### AGGREGATE (MARKET) DEMAND

$$Q(p) = q_1 + q_2 + \dots = D_1(p) + D_2(p) + \dots \text{ (add quantities across buyers at the same price)}$$

*Reads:* market demand = every individual's demand added up **at each price**. —  $Q(p)$  = total quantity the whole market wants at price  $p$  •  $q_1, q_2, \dots$  = the quantities buyer 1, buyer 2, ... each want at *that same* price •  $D_1(p), D_2(p)$  = each buyer's own demand function. (It's a **horizontal** sum — fix the price, add the quantities.)

*Deck example:* Mike's inverse demand  $p = 100 - 2q_M$ , Linda's  $p = 100 - \frac{1}{2}q_L \rightarrow$  invert each to  $q(p)$ , then add: at each price, total  $q = q_M + q_L$ . (Watch for a **kink** where one buyer drops out at high prices.)

- **The T-shirt trap (don't confuse products):** seeing pricey Gucci shirts outsell cheap generics is **not** an upward-sloping demand curve — those are **different goods**. A demand curve compares a good to **itself** (Gucci cutting *its own* price sells *more*). Real upward-sloping demand = the very rare **Giffen good** (Irish-famine potatoes; poor Chinese rice households — Jensen & Miller 2008).

## 2.2 Supply — the supply function

**Core:** supply answers "how much do firms bring to market at each price?" Firms are **price-takers** (they choose quantity, not the price). The curve usually slopes **up** — higher price  $\rightarrow$  more supplied.

- *In plain terms — why upward:* a higher price (i) makes it worth producing more even as costs rise, and (ii) draws in higher-cost producers who couldn't profit at the low price.
- **The supply function** (quantity supplied depends on price and input/other factors):

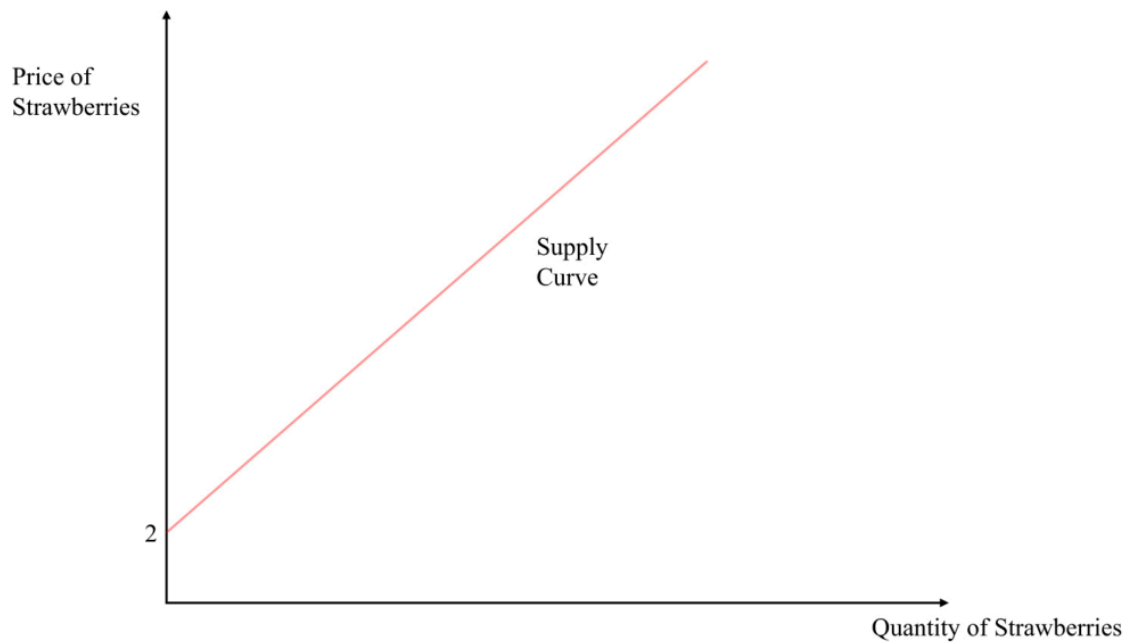
#### SUPPLY FUNCTION & ITS INVERSE

$$Q_S = S(p, p_o) \cdot \text{linear example: } Q_S = -2 + p \implies \text{inverse supply } p = 2 + Q_S$$

*Reads:* how much firms bring to market depends on the price they get and their costs. —  $Q_S$  = quantity supplied •  $S(\cdot)$  = "is a function of" •  $p$  = the good's **price** •  $p_o$  = price of **inputs** (wages, materials — higher costs shift supply in). In the example the "-2" is the price-intercept read backwards: **price must exceed 2 before any is supplied** (at  $p = 2, Q_S = 0$ ).

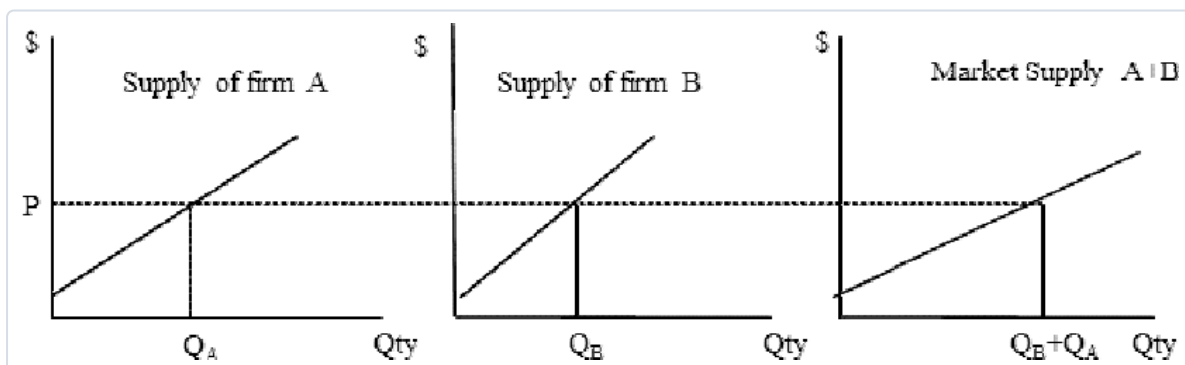
- **Why the "-2" (a negative intercept that isn't weird):** read it in **inverse** form — **price must clear €2 before any is supplied** (at  $p = 2, Q_S = 0$ ). *Check:*  $p = 3 \rightarrow Q_S = 1$ ;  $p = 4 \rightarrow Q_S = 2$ .

## Microeconomics



**How to read it (the supply curve).** Start at the **€2 price-intercept** (below it, zero supply) and climb: each extra euro of price brings one more litre to market. Like demand, an **own-price** change = **move along**; anything else = **shift**.

- **Supply shifters (move the whole curve):** input/cost  $\uparrow$  (wages, transport)  $\rightarrow$  shift **in** (need a higher price for the same quantity); **cheaper inputs / new tech / more sellers / market opens to foreign suppliers**  $\rightarrow$  shift **out**; **disruptions** (strikes, extreme weather, mine closures)  $\rightarrow$  shift **in**.
- **From individual to aggregate supply (deck):** market supply = **horizontal sum** of each firm's supply — add quantities across firms at each price.



**How to read it (aggregate supply).** At any price **P**, read each firm's quantity off its own curve ( $Q_A$ ,  $Q_B$ ) and **add them across** — the market curve is the sum, so it's **flatter** (more responsive) than any single firm's.

## ★ 2.3 Market equilibrium — solving for $p^*$ and $q^*$

**Core:** equilibrium = the **one price where quantity demanded = quantity supplied** ( $Q_D = Q_S$ ), the **market-clearing price** — no shortage, no glut, nobody wants to move. **This is the calculation to master.**

- The recipe (memorise these three steps):

STEP 1 — SET DEMAND = SUPPLY

$$Q_D = Q_S \Rightarrow 10 - p = -2 + p$$

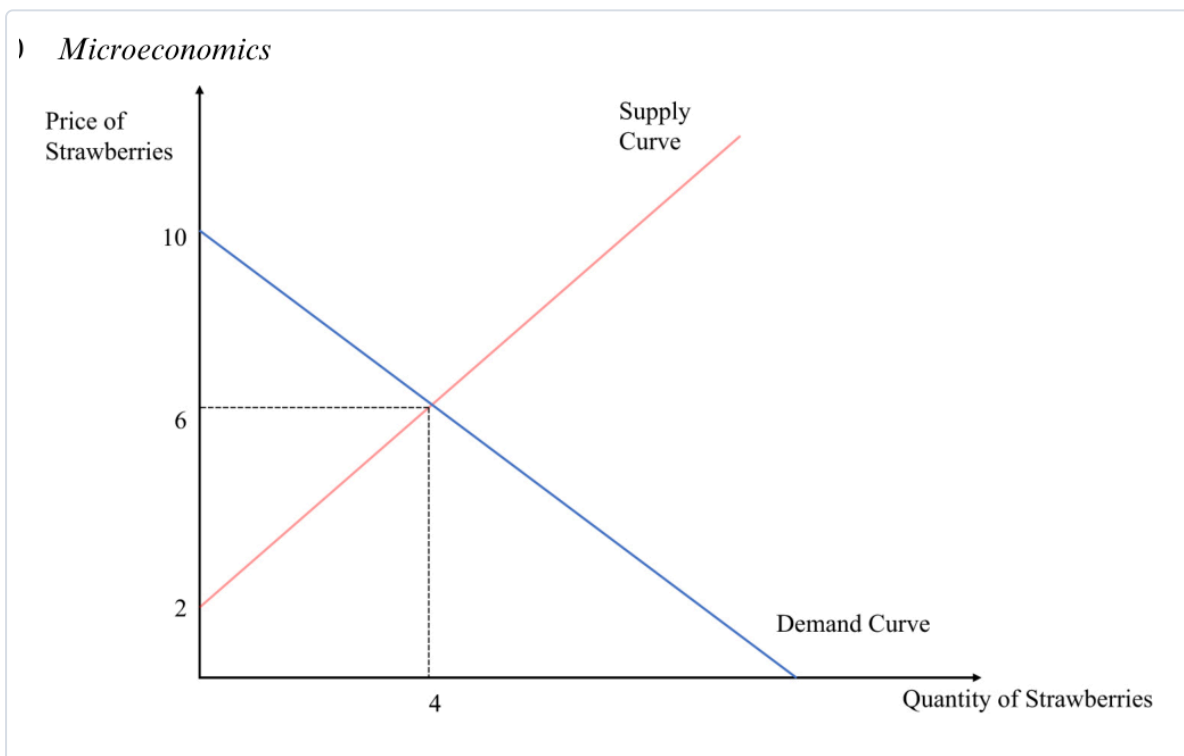
STEP 2 — SOLVE FOR THE EQUILIBRIUM PRICE  $P^*$

$$10 + 2 = p + p \Rightarrow 12 = 2p \Rightarrow p^* = 6$$

STEP 3 — PUT  $P^*$  BACK INTO EITHER CURVE FOR  $Q^*$

$$Q^* = 10 - 6 = 4 \text{ (check: } Q_S = -2 + 6 = 4 \checkmark) \Rightarrow \text{equilibrium } (p^*, q^*) = (6, 4)$$

*Reads:* find the price where the two plans agree, then the quantity that trades there. —  $Q_D$  = quantity demanded •  $Q_S$  = quantity supplied • setting  $Q_D = Q_S$  solves for  $p^*$  = the market-clearing **price**; substituting  $p^*$  back into either curve gives  $q^*$  = the **quantity** traded. The star (\*) just marks "equilibrium value." (Both curves must give the *same*  $q^*$  — that's the  $\checkmark$  check.)



**How to read it (equilibrium).** The **crossing point** is the only place the two plans agree — buyers want exactly what sellers offer (4 litres at €6). Everywhere else, one side is frustrated and price gets pushed toward the cross.

- **Off-equilibrium: excess demand & excess supply** (this is also *exactly* how you handle price controls below):

EXCESS DEMAND (SHORTAGE) – PRICE BELOW  $P^*$

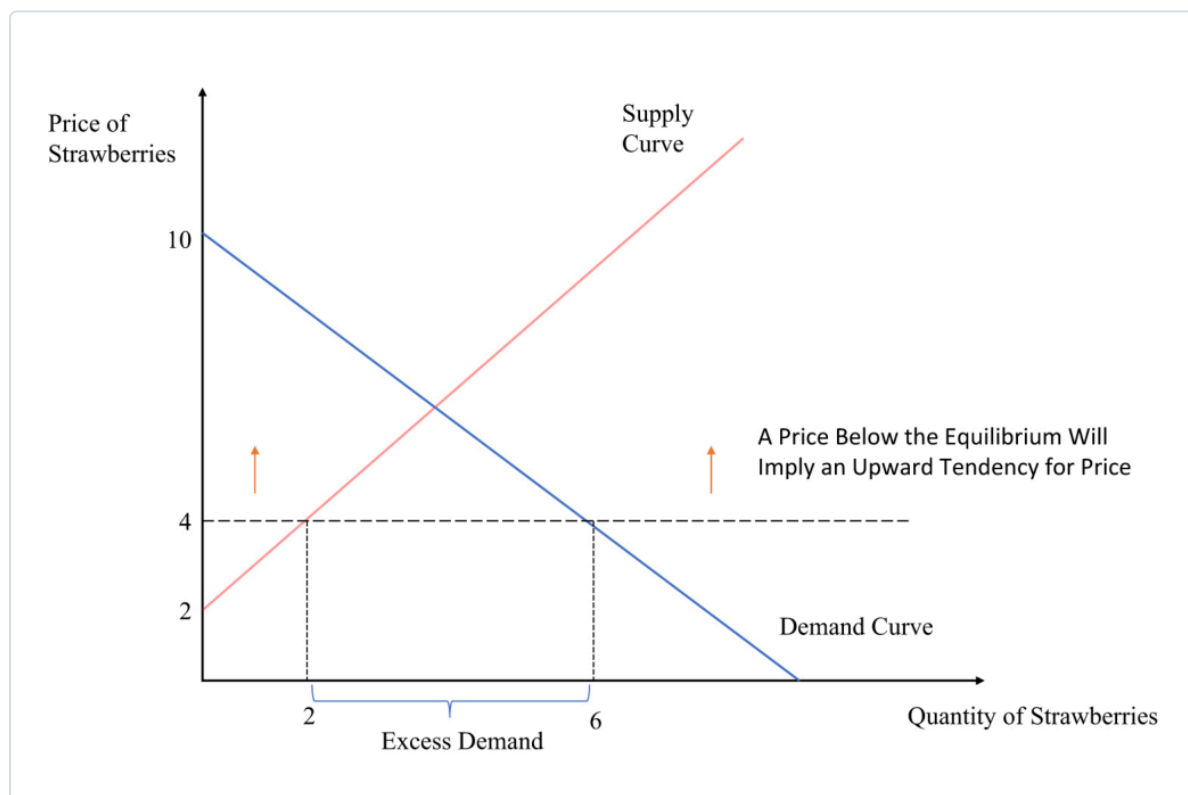
at  $p = 4$ :  $Q_D = 10 - 4 = 6$ ,  $Q_S = -2 + 4 = 2 \Rightarrow \text{shortage} = Q_D - Q_S = 4 \rightarrow$   
upward pressure on price

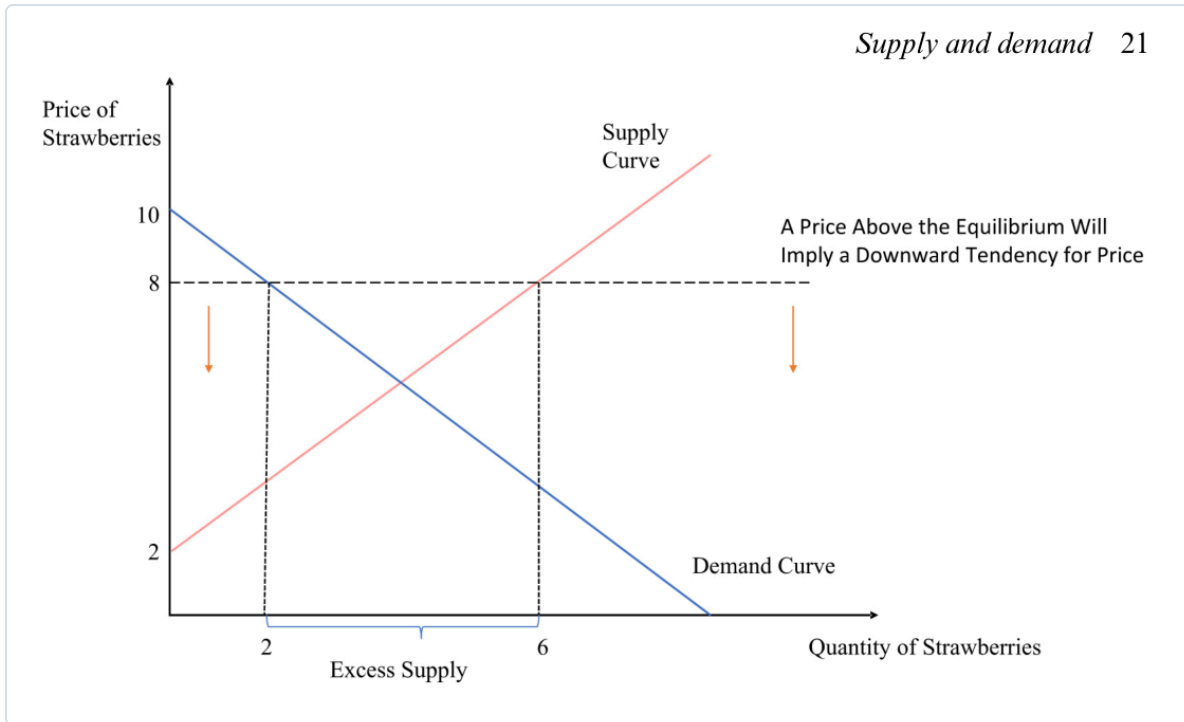
*Reads:* below equilibrium, buyers want more than sellers will supply. — plug the low price ( $p = 4$ ) into **both** curves  $\rightarrow Q_D = 6$  wanted,  $Q_S = 2$  offered; the gap  $Q_D - Q_S = 4$  is the **shortage**, which pushes the price **up** toward  $p^*$ .

EXCESS SUPPLY (SURPLUS) – PRICE ABOVE  $P^*$

at  $p = 8$ :  $Q_D = 10 - 8 = 2$ ,  $Q_S = -2 + 8 = 6 \Rightarrow \text{surplus} = Q_S - Q_D = 4 \rightarrow$   
downward pressure on price

*Reads:* above equilibrium, sellers offer more than buyers want. — plug the high price ( $p = 8$ ) into **both** curves  $\rightarrow Q_S = 6$  offered,  $Q_D = 2$  wanted; the gap  $Q_S - Q_D = 4$  is the **surplus**, which pushes the price **down** toward  $p^*$ .





**How to read them.** Pick the off-equilibrium price, drop a vertical line, and read **both** curves: the **horizontal gap** between them is the shortage (demand side wider) or surplus (supply side wider). The gap is what pushes price back toward  $p^*$ .

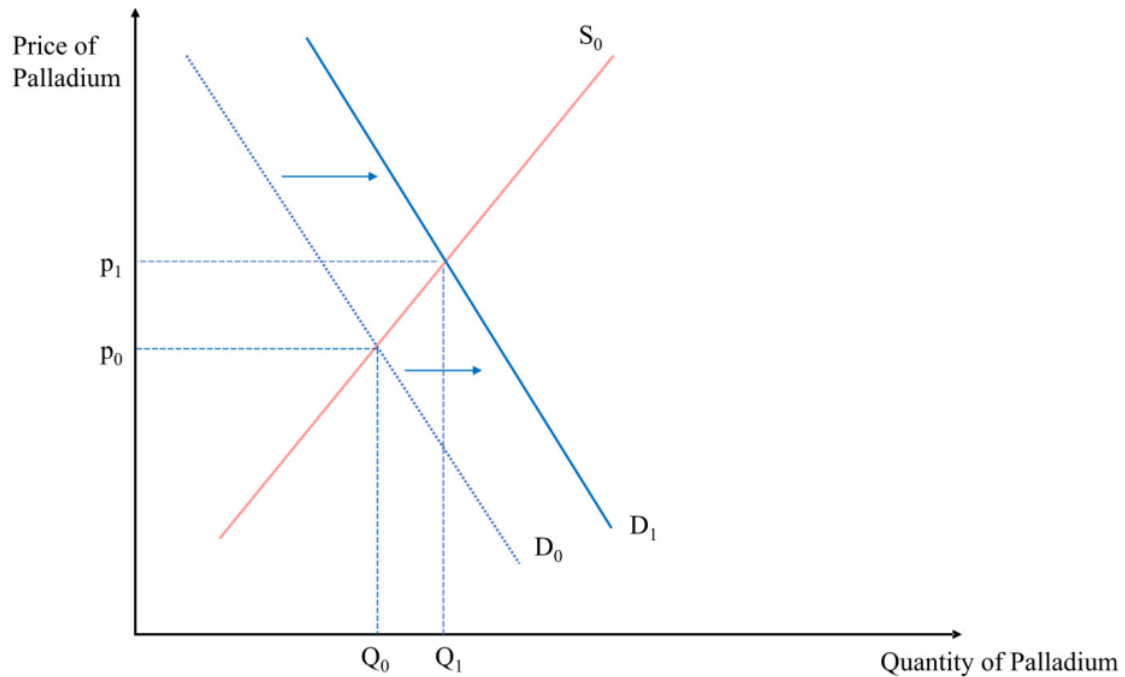
- **Marshall's scissors:** asking whether *demand or supply* sets the price is like asking *which blade of a scissors cuts* — **both do**. This dissolves the **diamond–water paradox**: water is cheap not because it's little wanted but because it's **abundant in supply**; diamonds are dear because they're **scarce**.
- (Who actually "sets" the price if everyone's a price-taker? The theoretical **Walrasian auctioneer** — and **Vernon Smith's 1962 experiments** showed real people reach the equilibrium price fast.)

## 2.4 Shifts in the equilibrium — comparative statics

**Core:** to analyse any shock, **don't jump to price/quantity**. First figure out **which curve moves and which way**; the effect on  $p^*$  and  $q^*$  follows.

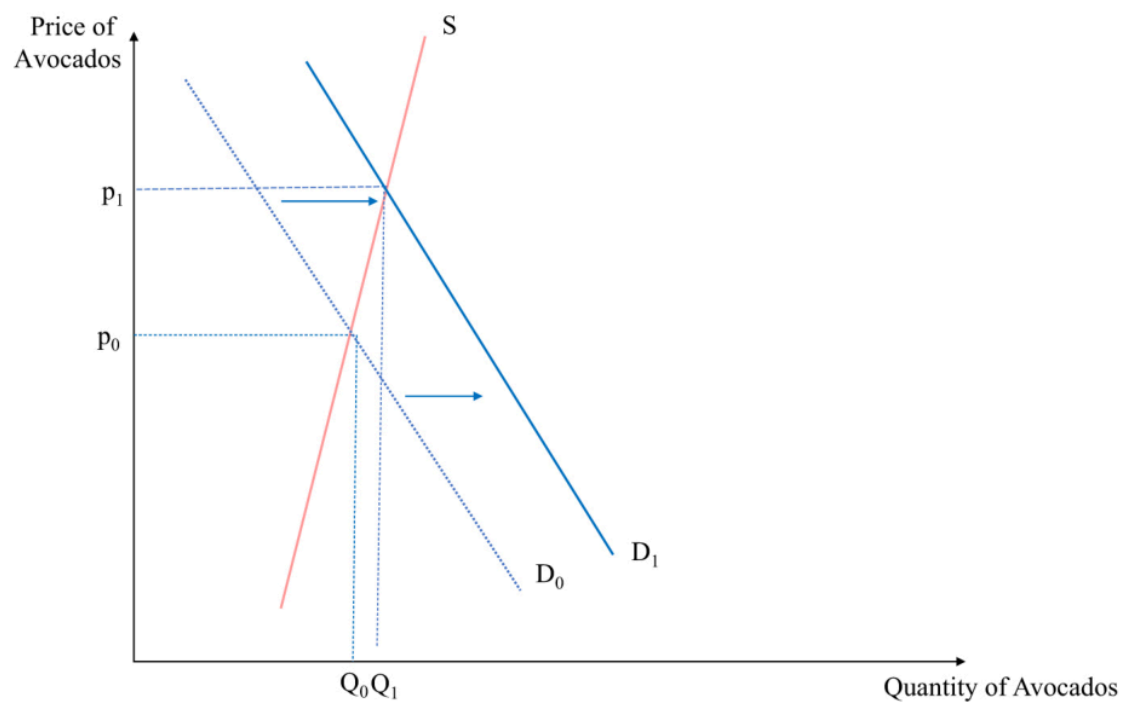
- **The method (Butera/Marshall — one change at a time, *ceteris paribus*):** ① Demand or supply? (falling incomes/other-good prices → demand; input costs/disruptions → supply). ② **Which direction** does it move *at a given price*? ③ Then read the new crossing. ④ **Reminder:** a *shift* is not a movement along — so a demand shift giving **both** higher price **and** higher quantity does **not** break the law of demand.

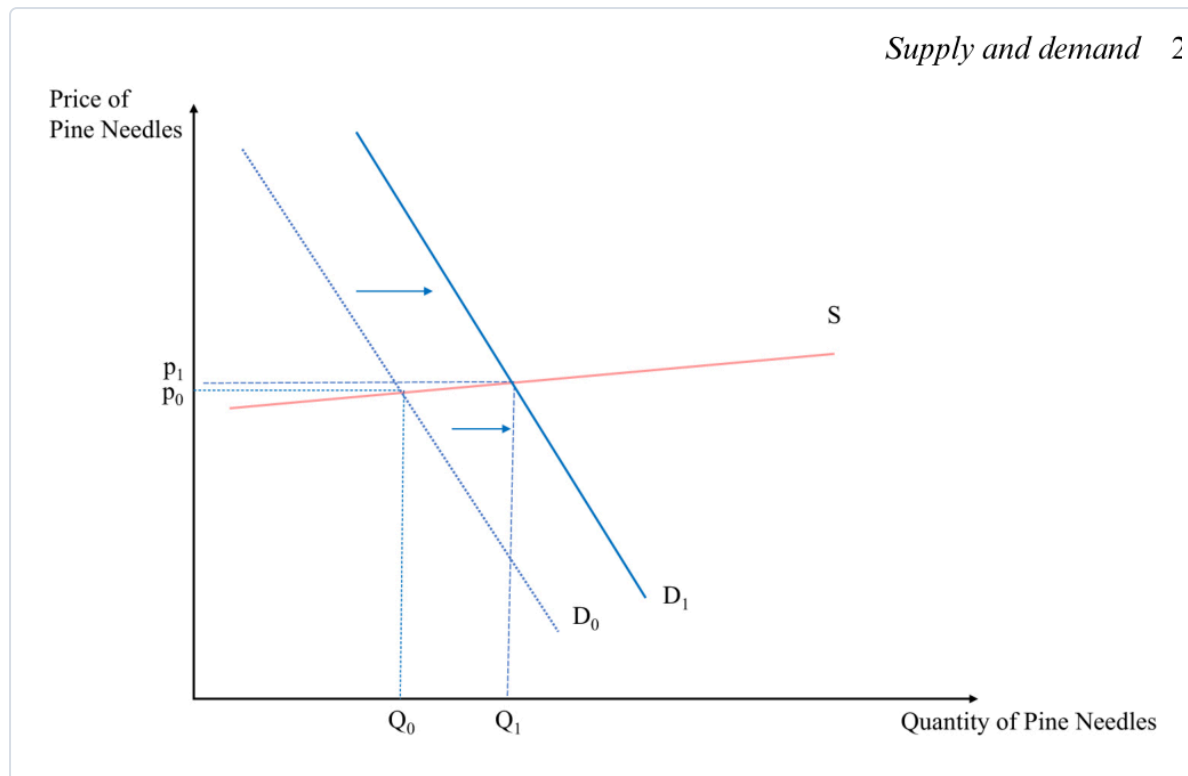
## Supply and demand



**How to read it (a demand shock).** Demand shifts **out** ( $D_0 \rightarrow D_1$ ); supply is unchanged, so producers **move along** supply to the new cross — **both  $p$  and  $q$  rise**. (Add a simultaneous supply **fall** and price rises *unambiguously*, but the **quantity effect becomes ambiguous** — it depends which shift is bigger.)

- ★ **Slope decides the *size* of the effect.** A demand shock's impact depends on the **slope of the supply curve**:





**How to read them.** Same outward demand shift, opposite results. **Steep supply** (avocados — trees take years, stock is fixed short-run): price jumps, quantity barely moves. **Flat supply** (pine needles — anyone can gather more): quantity jumps, price barely moves. *The flatter the other curve, the more a shock shows up as quantity; the steeper, the more it shows up as price.*

## Price ceilings & floors — when the government fixes the price (deck)

**Core:** a **binding price control** stops the market clearing, and you find the resulting gap with the **exact excess-demand/supply calculation above** — just set  $p$  to the controlled price.

- **Price ceiling** = a legal **maximum** (e.g. **rent control**). To bind it must be **below  $p^*$** , so  $Q_D > Q_S \rightarrow$  a **shortage** (queues, waiting lists, black-market resale). *Compute:* plug the capped  $p$  into both curves; shortage =  $Q_D - Q_S$ .
- **Price floor** = a legal **minimum** (e.g. the **minimum wage** in the labour market). To bind it must be **above  $p^*$** , so  $Q_S > Q_D \rightarrow$  a **surplus** (unsold goods; in the labour market, **unemployment** — more people want to work than firms will hire). *Compute:* surplus =  $Q_S - Q_D$ .
- **The trade-offs (positive, not normative): ceilings** — **pro:** keep essentials affordable for the less-well-off; **con:** shortages, quality decline, and markets "find a way around" (rent control can even be **regressive**; Venezuela's caps produced empty shelves). **floors** — **pro:** a living wage; **con:** firms substitute **capital for labour** (self-checkout at Føtex), cut hours, or make jobs temporary. "Good economics generates **positive** statements — evidence on costs and benefits — to inform the **normative** political choice."

## Is the model realistic? — perfect competition (deck)

**Core:** the supply-demand model assumes **perfect competition** — "every model is wrong, but useful."

- **Assumptions:** ① every agent is a **price-taker**; ② goods are **homogeneous** (perfect substitutes — one farm's wheat = another's; Shell petrol = Exxon petrol); ③ **perfect information** about prices/quality; ④ **low transaction costs**; (and many buyers & sellers). **Break these** and you need refinements — **monopoly, market failures** (later chapters).

## Cases & examples

Each: *the example* → *the concept it teaches*.

- **Strawberries (the running example)** → the whole toolkit:  $Q_D = 10 - p$ ,  $Q_S = -2 + p$ , equilibrium (6, 4), and the €4/€8 shortage/surplus.
- **Diamond–water paradox** → price is set by **supply and demand** (Marshall's scissors), not usefulness — water's cheap because it's abundant.
- **Palladium** (tighter emission rules → catalytic-converter demand ↑; S. African mine disruptions → supply ↓) → **comparative statics** with one and two curves moving.
- **2008 world rice spike & Haiti riots** (India's export ban cut supply; panic stockpiling shifted demand out) → a real **double shift**; prices later fell as stockpiles released + the financial crisis hit demand.
- **Instant noodles** → **inferior good**; **raspberries** → **substitute**; **cereal & milk** → **complements**.
- **Giffen goods** (Irish-famine potatoes; poor Chinese rice households) → the rare **upward-sloping demand** exception.
- **COVID toilet-paper stockpiling** → an **expectations** shift of demand.
- **Avocados vs Swedish pine-needle tea** → **steep vs flat supply** sizing a demand shock.
- **Rent control / Venezuela** → **price ceiling** → **shortage**; **minimum wage** → **price floor** → **surplus** (unemployment).

## Formula sheet — quick reference

| Object                    | Formula                                                                  | Note                                     |
|---------------------------|--------------------------------------------------------------------------|------------------------------------------|
| Demand function           | $Q_D = D(p, p_s, p_c, Y, \tau)$                                          | many variables; curve fixes all but p    |
| Linear demand / inverse   | $Q_D = a - b \cdot p / p = (a - Q_D)/b$                                  | a = intercept (shifters), b = slope      |
| Supply function / inverse | $Q_S = -2 + p / p = 2 + Q_S$                                             | negative intercept = min price to supply |
| Aggregate (market)        | $Q(p) = \sum D_i(p) ; \sum S_i(p)$                                       | <b>horizontal</b> sum (add quantities)   |
| <b>Equilibrium</b>        | <b>set <math>Q_D = Q_S \rightarrow p^*</math>, then <math>q^*</math></b> | the market-clearing point                |
| Excess demand (shortage)  | $Q_D - Q_S$ at $p < p^*$                                                 | → upward price pressure; price ceiling   |
| Excess supply (surplus)   | $Q_S - Q_D$ at $p > p^*$                                                 | → downward pressure; price floor         |

## Exam pointers

- **Solve a market from scratch:** given a linear demand and supply, **set them equal**, get  $p^*$ , back-substitute for  $q^*$  — the strawberry case ( $p^*=6$ ,  $q^*=4$ ) is the model answer.

- **Compute a shortage/surplus:** plug a below- or above-equilibrium price into *both* curves and take the gap ( $Q_D - Q_S$  or  $Q_S - Q_D$ ) — and know a **ceiling**→shortage, **floor**→surplus.
- **Diagnose movement vs shift**, and for a shock say **which curve moves, which way**, and the effect on  $p^*$  &  $q^*$  (incl. the **ambiguous-quantity** double-shift and the **slope** point).
- Explain **why demand slopes down** (scarcity + diminishing marginal utility) and **Marshall's scissors** / the diamond–water paradox.
- Name the **perfect-competition** assumptions and why the model is still "useful though wrong."

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## ★ Fun facts & memorable details (Lecture 2)

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Sticky bits from Supply & Demand.

- **Marshall's scissors:** asking whether demand or supply sets the price is like asking **which blade of the scissors** does the cutting — both do.
  - The **diamond–water paradox** is only a paradox if you forget supply: water is life-or-death yet cheap **because it's abundant**.
  - **Giffen goods** really exist: when their staple got dearer, poor Chinese rice households bought **more** rice, not less (Jensen & Miller 2008) — they could no longer afford anything else.
  - A negative supply intercept ( $Q_S = -2 + p$ ) isn't a mistake — it just means **price must top €2 before anyone supplies a drop**.
  - **Vernon Smith** won a Nobel (2002) for showing that real people in a lab **converge on the equilibrium price** startlingly fast — the "Walrasian auctioneer" made flesh.
  - Rent control's dirty secret: it can be **regressive** and spawn black-market "key money" — a "tale of good intentions" (cf. Venezuela's empty shelves).
  - Minimum wage as a price floor nudges firms toward **capital over labour** — the **self-checkout at Føtex** is a supply-and-demand diagram in the wild.
-